

Small Models, Big Reactions: Studying Efficient Emoji Prediction

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October 7, 2026

1 Problem and Motivation

Emoji are widely used in online communication to express emotion, tone, humor, and reactions that may not be obvious from text alone. We study emoji prediction as a multi-class text classification task: given a short social-media post with its emoji removed, the model predicts one of a fixed set of emoji classes.

Prior work has shown that both simple classifiers and pretrained language models can perform this task, but these approaches vary greatly in predictive performance and computational cost. Understanding this tradeoff is useful because a smaller model may be preferable when computational resources or inference speed are limited, even if it sacrifices some predictive performance. We therefore ask: **What is the tradeoff between model complexity and emoji-prediction performance?** Using the TweetEval emoji benchmark, we will compare models of increasing complexity, including TF-IDF + logistic regression, a BiLSTM classifier, DistilBERT, and a larger Transformer such as BERT-base or RoBERTa-base. We will evaluate macro-F1, parameter count, and inference latency, and analyze frequently confused emoji classes. This study will help us understand whether the performance gains of larger models justify their additional computational cost, and whether those gains are concentrated in difficult or semantically similar emoji classes.

2 Literature Survey

Emoji prediction has been studied as a social-media text classification task in prior work. The SemEval-2018 Task 2 benchmark formalized the problem as predicting the most likely emoji associated with a tweet and evaluated systems using macro-F1 because of the imbalance among emoji classes [2]. A wide range of approaches were explored in the shared task, including bag-of-words classifiers, support vector machines, convolutional neural networks, recurrent neural networks, and attention-based models. Some relatively simple systems were surprisingly competitive, while more complex neural models attempted to capture semantic and contextual information more effectively.

Later, TweetEval incorporated emoji prediction into a standardized benchmark for Twitter -specific classification tasks and provided fixed training, validation, and test splits to-

gether with stronger pretrained language-model baselines [1]. TweetEval showed that pretrained Transformer models can improve performance, but emoji prediction remains difficult because the label distribution is highly skewed and several emoji classes have similar meanings. These results motivate our study of whether the additional complexity of larger models produces enough improvement to justify their higher computational cost.

3 Data Requirements

We will use the TweetEval emoji classification benchmark [1], which contains short social-media posts labeled with one of 20 emojis. The provided splits contain approximately 45,000 training, 5,000 validation, and 50,000 test examples. The dataset directly matches our task and enables comparison with established baselines.

TweetEval contains substantial class imbalance and semantically similar emoji classes, making macro-F1 and class-level error analysis important. We will preserve the provided splits and apply only limited preprocessing, such as removing residual Unicode artifacts when necessary.

4 Experiment Plan

We will compare several models of increasing complexity on the TweetEval emoji prediction task. Our baseline will be TF-IDF features with logistic regression. We will then compare this baseline with a BiLSTM classifier, DistilBERT, and a larger pretrained Transformer such as BERT-base or RoBERTa-base. All models will use the same train, validation, and test splits to ensure a fair comparison.

We will evaluate predictive performance using macro-F1, which is appropriate for the imbalanced emoji classes. To measure computational cost, we will record parameter count and inference latency under the same hardware and batch settings. We will also perform class-level and qualitative error analysis using confusion matrices and representative misclassified examples, focusing on whether larger models better distinguish semantically similar emoji classes.

5 Feasibility and Next Steps

We will run experiments in Google Colab Pro, using an NVIDIA T4 GPU as our expected resource and an A100 when available. Our first experiment will establish the TF-IDF + logistic regression baseline and verify the TweetEval training and evaluation pipeline. By the milestone, we aim to evaluate the baseline, BiLSTM, and at least one pretrained Transformer, with preliminary macro-F1, model-size, and inference-latency results. By the final report, we will complete the model comparison and qualitative error analysis.

Potential challenges include GPU availability, class imbalance, noisy social-media text, and preprocessing artifacts. If compute becomes limiting, we will reduce the number of larger Transformer models while preserving the performance-complexity comparison.

References

- [1] Francesco Barbieri, Jose Camacho-Collados, Luis Espinosa-Anke, and Leonardo Neves. TweetEval: Unified benchmark and comparative evaluation for tweet classification. In *Findings of the Association for Computational Linguistics: EMNLP 2020*, pages 1644–1650, Online, 2020. Association for Computational Linguistics.
- [2] Francesco Barbieri, Jose Camacho-Collados, Francesco Ronzano, Luis Espinosa-Anke, Miguel Ballesteros, Valerio Basile, Viviana Patti, and Horacio Saggion. SemEval 2018 task 2: Multilingual emoji prediction. In *Proceedings of the 12th International Workshop on Semantic Evaluation*, pages 24–33, New Orleans, Louisiana, 2018. Association for Computational Linguistics.